HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **SystemVerilog** testbench components, adhering to **UVM methodology** for complex processor designs.
- Created and executed directed and constrained-random verification test cases, validating **hardware functionality** against architectural specifications.
- Analyzed simulation results and debugged functional issues in collaboration with senior verification and design engineers.
- Contributed to the improvement of verification automation scripts and infrastructure using **Python**, reducing test execution time by **18%**.
- Collected and analyzed **functional and code coverage metrics**, identifying gaps and driving verification closure for embedded processor cores.
- Participated actively in **design reviews** and verification strategy discussions, providing input on potential hardware implementation issues.
- Containerized verification environments using **Docker** and deployed them to a shared **Linux**-based cluster for consistent test execution.

SKILLS

Hardware Verification – SystemVerilog, UVM, Constrained-Random Verification, Functional Coverage, Code Coverage, Assertions

Programming Languages – Python, C++, C

Hardware Description Languages – Verilog, VHDL

Computer Architecture – Processors, DSP, AI Accelerators, SoCs, Microprocessors, Digital Logic Design

Tools & Methodologies – Simulation Tools, Debugging Tools, Scripting, Linux, Git, Jenkins, Docker, CI/CD

Methodologies – Agile, Scrum

PROJECTS

DSP AI Accelerator Verification Environment

- Developed a **SystemVerilog/UVM** testbench for a novel DSP AI accelerator, focusing on verifying core arithmetic and data-path operations.
- Implemented directed tests and a constrained-random stimulus generator to cover a wide range of operational modes and edge cases.
- Integrated **Python** scripting for automated test execution and result parsing, streamlining the verification workflow for the team.

Processor Core RTL Simulation and Debug

- Assisted in verifying a complex processor core design by creating **SystemVerilog** test cases and analyzing simulation logs.
- Debugged functional failures by correlating simulation waveforms with RTL code and design specifications, identifying issues related to instruction decoding.
- Contributed to coverage analysis, reporting on the state of functional coverage attainment for key processor features.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)